# Specifications

## CFD Project

### Aerodynamic Modelling of the Drafting Effect in NASCAR

*Translating moving meshes with STAR-CCM+*

**Group 4**

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# 1. Project objectives

— **Main objective:** demonstrate that STAR-CCM+ can accurately simulate the drafting effect during a NASCAR overtake using a translating moving mesh.
— **Secondary objective:** analyse how the distance between vehicles affects their aerodynamic coefficients (drag, lift) and related effects such as side-drafting.

# 2. Static modelling

## 2.1 Geometry

Description of the vehicle geometry:

— Highly simplified body profile with rear wing
— Length: 5 m
— Height: 1.42 m
— Width: 1.95 m
— Model: Chevrolet Camaro Gen-6

## 2.2 Mesh

— Type: mixed hexahedral mesh
— Growth rate: 1.3
— Boundary layer: 8 prism layers

## 2.3 Boundary conditions

— Inlet: wind inlet velocity = 80 m/s, turbulence intensity = 1 %
— Outlet: outlet pressure = 1.013 bar
— Walls: no-slip conditions on the vehicles and the ground
— Symmetry: symmetry planes (to be specified)

## 2.4 Computing the drag coefficients

For each longitudinal spacing $x/L$ between the following car and the lead car, the drag coefficient $C_{D,2}(x/L)$ of the second car is computed as:

$$C_{D,2}(x/L) = \frac{2\,F_{D,2}}{\rho\,V_\infty^2\,A}$$

where:

— $F_{D,2}$ is the drag force measured on the following vehicle,
— $\rho$ the air density (kg/m$^3$),
— $V_\infty$ the free-stream inlet velocity (m/s),
— $A$ the reference area (m$^2$).

$C_{D,1}$ (lead car) and $C_{D,2}(x/L)$ (following car) will be plotted against $x/L$ for comparison.

## 2.5 Comparing velocities

For each configuration, the mean velocity $\overline{V}_i$ around each vehicle $i = 1, 2$ is extracted. The velocity ratio is defined as:

$$R_V(x/L) = \frac{\overline{V}_2(x/L)}{\overline{V}_1}.$$

A plot of $R_V$ against $x/L$ will highlight the speed gain due to drafting.

## 2.6 Velocity and pressure fields

**Velocity fields** 2D cuts in the longitudinal plane show the streamlines coloured by velocity magnitude. These visualisations identify the drafting pocket and its extent.

**Pressure fields** The surface pressure $p$ on the rear wing of each car is compared. The rear-axle downforce is given by:

$$F_{p,i} = \iint_{S_{ar,i}} p\,\hat{n} \cdot \hat{z}\,\mathrm{d}S,$$

where $S_{ar,i}$ is the rear-wing surface of vehicle $i$. The local lift coefficient $C_{L,ar,i}$ is obtained from:

$$C_{L,ar,i} = \frac{F_{p,i}}{\frac{1}{2}\,\rho\,V_\infty^2\,A_{ar}},$$

with $A_{ar}$ the rear-wing area.

## 2.7 Recirculation zone behind the following vehicle

For each spacing $x/L$, the recirculation zone is characterised by:

— **Recirculation bubble length** $L_{rec}(x/L)$**:** distance between the rear edge of the following vehicle and the point where the axial velocity becomes positive again.
— **Recirculation volume** $V_{rec}(x/L)$**:** integral of the volume where the axial velocity $u_x < 0$.
— **Vortex intensity:** maximum vorticity amplitude $\omega_{max}$ in the bubble.

These quantities are extracted from the velocity field and plotted against $x/L$. The analysis covers:

1. The evolution of $L_{rec}$: correlation with the drag reduction $C_{D,2}$.
2. The variation of $V_{rec}$: impact on the rear stability of the vehicle.
3. The vortex intensity: indication of possible lateral disturbances during the overtake.

## 2.8 Summary of comparisons

— **Drag:** evolution of $C_{D,1}$ and $C_{D,2}(x/L)$.
— **Velocity:** ratio $R_V(x/L)$ and analysis of the speed gain.
— **Rear downforce:** comparison of $C_{L,ar,2}(x/L)$ vs $C_{L,ar,1}$.
— **Recirculation:** correlations between $L_{rec}$, $V_{rec}$ and aerodynamic performance.

The sections above will be completed with data tables and plots from the CFD simulations.

# 3. 2D dynamic modelling

## 3.1 Overset moving mesh

The dynamic modelling uses the STAR-CCM+ Overset technique to simulate the relative motion of the vehicles:

- **Moving zone (Overset):** region around the following vehicle, moving at constant or variable speed
- **Background zone:** static mesh of the main domain containing the lead vehicle
- **Overlap interface:** automatic conservative interpolation between the two zones
- **Time step:** $\Delta t = 0.002\,\mathrm{s}$, CFL criterion $< 1$ in critical zones

## 3.2 Computing the dynamic drag coefficients

For each initial spacing $x/L$ and under transient conditions, we compute:

$$C_{D,2}^{dyn}(t) = \frac{2\,F_{D,2}(t)}{\rho\,V_\infty^2\,A}$$

The time-averaged value is extracted over the drafting phase:

$$\langle C_{D,2}^{dyn} \rangle = \frac{1}{T}\int_0^T C_{D,2}^{dyn}(t)\,\mathrm{d}t$$

## 3.3 Velocity evolution over time

The instantaneous velocity ratio is defined as:

$$R_V^{dyn}(t) = \frac{V_2(t)}{V_1(t)}$$

The analysis covers:

- The acceleration of the following vehicle as it enters the drafting pocket
- The stabilisation of the speed gain during the sustained drafting phase
- The deceleration when leaving the draft (the "wall of air" effect)

## 3.4 Dynamic velocity and pressure fields

**Velocity fields** Animations showing the time evolution of:

- Streamlines in the longitudinal plane
- Velocity magnitude contours
- The moving drafting pocket

**Pressure fields** Time evolution of:

- Pressure distribution on the vehicle surfaces
- Instantaneous aerodynamic downforce on the wings
- Pressure gradient between the lead and following vehicles

### 3.5 Dynamic recirculation zone

At each time $t$, we measure:

— $L_{rec}(t)$: instantaneous length of the recirculation bubble behind the following car
— $V_{rec}(t)$: volume where the axial velocity $u_x < 0$
— $\omega_{max}(t)$: maximum vorticity in the bubble

The analysis examines:

1. Fluctuations of $L_{rec}$ during the approach and drafting phases
2. Correlation between variations of $V_{rec}$ and vehicle instabilities
3. Dominant frequency of the $\omega_{max}$ oscillations (spectral analysis)

### 3.6 Validation of the 2D dynamic model

Validation criteria:

— **Mass conservation:** error $< 0.1\,\%$ at every time step
— **Consistency with static results:** $\langle C_D^{dyn} \rangle \approx C_D^{stat}$ for equivalent positions
— **Reproducibility:** deviation $< 2\,\%$ between repeated simulations
— **Time convergence:** $\Delta t$ refined until the results stabilise

## 4. 3D dynamic modelling

### 4.1 3D Overset set-up

Extension of the Overset technique to three dimensions:

— **Overset region:** box enclosing the following vehicle with sufficient margins
— **Background region:** main domain with the lead vehicle held fixed
— **Motion:** longitudinal and/or lateral translation depending on the scenario:
  — Axial approach: $x(t)$ variable, $y = 0$
  — Lateral overtake: $x$ fixed, $y(t)$ variable
  — Combined manoeuvre: $x(t)$ and $y(t)$ variable
— **Interpolation:** second-order conservative method

### 4.2 Dynamic scenarios

#### 4.2.1 Approach phase

— Initial position: $x/L = 3.0$, $y/W = 0$
— Following-car speed: $V_2 = V_1 + \Delta V$
— Simulation duration: until $x/L = 0.25$ (entry into the drafting zone)

#### 4.2.2 Sustained drafting phase

— Position: $x/L = 0.5$, $y/W = 0$

— Formation held
— Analysis of stability and fluctuations

## 4.3 Time analysis of forces

For each scenario, extraction and analysis of:

$$F_D(t), \quad F_L(t), \quad M_z(t) \qquad \text{(drag, lift, yaw moment)}$$

— Instantaneous evolution of the forces on both vehicles
— Identification of load peaks during transitions
— Spectral analysis (FFT) to identify characteristic frequencies
— Cross-correlation between the forces on vehicles 1 and 2

## 4.4 Evolution of flow structures

Animated visualisations showing:

— Deformation of the drafting pocket during the approach
— Vortices appearing and disappearing during the lateral motion
— Wake interaction during the overtake
— Flow reattachment after the vehicles separate

## 4.5 Validation of the 3D dynamic model

### 4.5.1 Numerical criteria

— Mass conservation at every time step: error $< 0.1\,\%$
— Field continuity at the Overset interface: consistent gradients
— Time-step independence: $\Delta t$ halved, convergence of force peaks
— Numerical stability: no non-physical oscillations

### 4.5.2 Physical consistency

— Comparison with 3D static results for frozen positions
— Compliance with global conservation laws (momentum, energy)
— Characteristic times consistent with the flow scales ($L/V_\infty$)
— Physically sensible trends: acceleration in the draft, deceleration when pulling out

### 4.5.3 Comparison with NASCAR data

— Speed gains observed in races: compatible order of magnitude
— Optimal drafting distances: consistent with driver practice
— Limitation: no precise telemetry data available for direct quantitative validation

## 4.6 Summary and recommendations

### 4.6.1 2D vs 3D comparison

— Vortex structures: simplified in 2D
— Aerodynamic coefficients: difference between 2D and 3D
— Computational cost: difference between 2D and 3D

### 4.6.2 Static vs dynamic comparison

— Transient phenomena: captured only in dynamic runs
— Mean values: consistency between static results and dynamic averages
— Load peaks: absent in static runs, critical for design
— Computation time

### 4.6.3 Optimal overtaking strategies

Summary of results for driver recommendations:

— Optimal drafting position
— Maximum speed gain
— Recommended overtaking line
— Zone to avoid: “drag bubble”

# 5. Conclusion and outlook

## 5.1 Objectives achieved

— STAR-CCM+ shown to be capable of simulating NASCAR drafting
— Overset moving meshes working and validated
— Influence of spacing quantified: $C_D$ map established
— Key physical phenomena identified and explained

## 5.2 Limitations and possible improvements

### 5.2.1 Current limitations

— Wheel rotation neglected: estimated impact $< 5\,\%$ on overall $C_D$
— Simplified underbody: affects lift accuracy
— Idealised atmospheric conditions: crosswind and atmospheric turbulence not modelled
— Multi-vehicle interaction: limited to 2 cars (packs not studied)

### 5.2.2 Outlook

— Modelling wheel rotation (Sliding Mesh, MRF)
— CFD / vehicle-dynamics coupling to simulate a complete manoeuvre

— Extension to multi-car formations (3–4 cars)
— Body shape optimisation with genetic algorithms coupled to CFD
— Experimental validation in a wind tunnel with motorised scale models